



**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA

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**FACULTY OF COMPUTING**

**SEMESTER 2 2025/2026**

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**SECP3843 – SPECIAL TOPIC IN DATA ENGINEERING**

**SECTION 1**

**TUTORIAL GAI**

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## **TABLE OF CONTENT**

|                                                          |          |
|----------------------------------------------------------|----------|
| <b>1.0 What is AI-assisted data engineering?</b>         | <b>3</b> |
| <b>2.0 What Claude/AI agent is used in the tutorial?</b> | <b>3</b> |
| <b>3.0 Steps to do prompt and coding</b>                 | <b>4</b> |
| <b>4.0 Reflection</b>                                    | <b>8</b> |
| 4.1 Nurul Asyikin Binti Khairul Anuar                    | 8        |
| 4.2 Lubna Al Haani Binti Radzuan                         | 9        |
| 4.3 Anis Safiyya Binti Janai                             | 9        |

## **1.0 What is AI-assisted data engineering?**

AI-assisted data engineering is the practice of using AI tools, particularly generative AI, to support and enhance data engineering tasks such as writing SQL transformations, generating tests, and troubleshooting failures. It shifts manual workflows toward more intent-driven development, where engineers focus on defining requirements while AI assists in implementation, optimization, and understanding of data systems (Gault, 2025).

It also improves productivity by reducing repetitive tasks and speeding up processes such as debugging pipeline failures, optimizing queries, and interpreting complex data structures (Chubb, 2025). In addition, AI can enhance data quality by suggesting validation rules and detecting anomalies while also improving data discovery through better understanding of schemas and data lineage (Averageguymedianow, 2025).

For example, in this tutorial, Express.dev was used to build a simple data pipeline that ingests weather data using an API from WeatherAPI and loads it into Snowflake at 15-minute intervals. The tutorial shows how AI-assisted data engineering can be applied to help design and automate parts of the pipeline, such as generating ingestion logic and helping with configuration decisions. Thus, AI is not a replacement for data engineers but rather a helper for decision-making, where humans still own correctness, governance, and business alignment (Gault, 2025).

## **2.0 What Claude/AI agent is used in the tutorial?**

In this tutorial, Express.dev, a platform developed by Nexla, provides an AI agent to help build and manage the entire pipeline for extract, transform, and load (ETL). Through a chat-based interface, the AI Agent allows the user to develop a pipeline using natural language prompts without manual coding or complex installation. During the tutorial, the AI agent guided the pipeline creation process by requesting information such as the WeatherAPI credentials and Snowflake connection settings before automatically configuring the required components.

The AI agent was used to automate several data engineering tasks that would need technical skills and programming expertise. The Express.dev creates the source connection, field mapping, and workflow settings for the requested data collection from a public API, which is WeatherAPI, and saves it to the Snowflake data warehousing platform. The AI agent also generates a schedule to execute the pipeline every 15 minutes. As shown in the tutorial, most of the pipeline setup process could be completed through conversation with the AI agent, reducing the amount of manual configuration required.

Furthermore, the AI agent also shows the agent's ability to perform data transformation. The AI agent developed a transformation phase and automatically wrote Python code when a new requirement to convert temperature values from Celsius to Fahrenheit was submitted. The updated workflow was then integrated into the existing pipeline without manual programming. This shows that AI can simplify pipeline development, but users still need to verify the results.

### 3.0 Steps to do prompt and coding

The tutorial began with prompting the idea of the pipeline that is to be created, as in Figure 1. The AI agent was instructed to create a data pipeline that retrieves current weather information for London at a specified interval of 2 minutes. The prompt identified the weather variables to be extracted, such as city name, temperature in Celsius, humidity, weather status, and the timestamp of the most recent update.

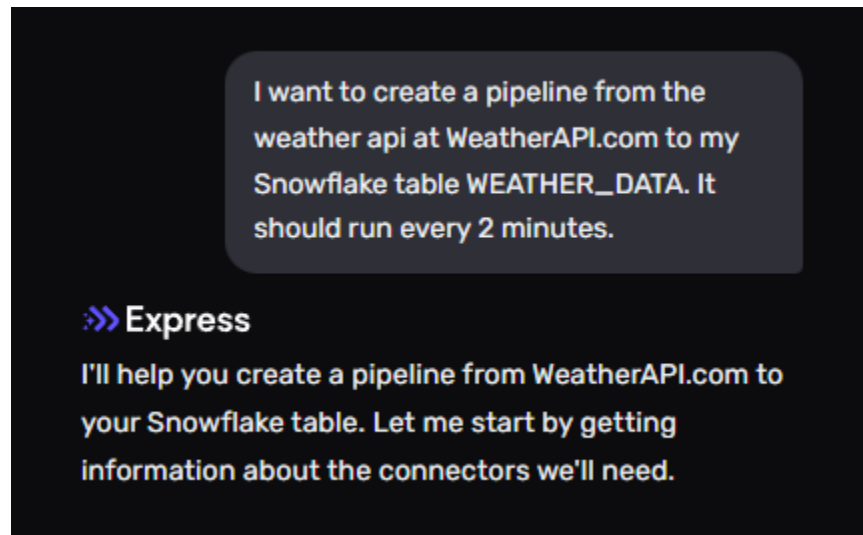


Figure 1: Prompting in Nexla Express.dev

[WeatherAPI.com](https://weatherapi.com) was used to configure the data source within the Nexla Express.dev platform. The connection for the weather service and the ETL pipeline was established with the API key generated from [WeatherAPI.com](https://weatherapi.com). After that, the incoming weather data were stored in the prepared destination tables called WEATHER\_DATA.

The connection to the Snowflake data platform was established using the appropriate credentials, database, schema, and table information. The AI agent will then automatically generate the sink configuration and map the extracted fields from the WeatherAPI source to their appropriate columns in the Snowflake table, which includes the mapping of current.temp\_c to TEMPERATURE, current.humidity to HUMIDITY, current.condition.text to WEATHER, and current.last\_updated to TIMESTAMP. These complete steps on creating a pipeline are shown in Figure 2.

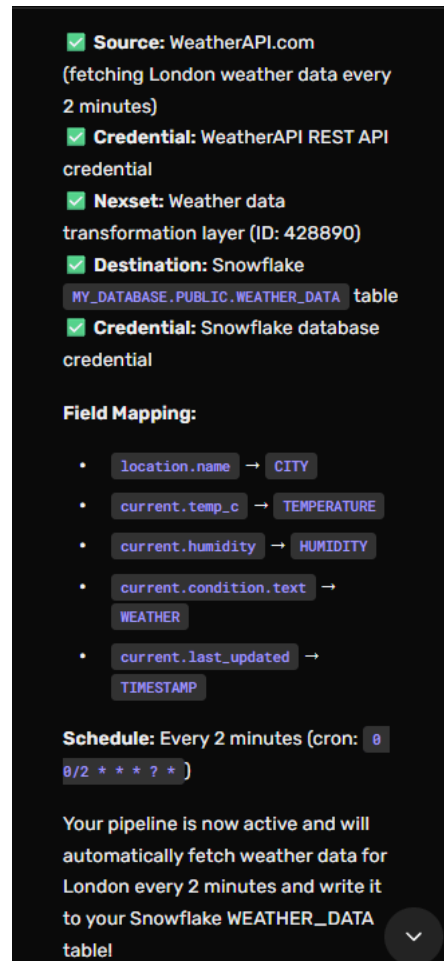


Figure 2: Complete steps to develop the pipeline

In addition, an extra column called TRANS\_TEMP\_F was added to the Snowflake table to store the temperature values in Fahrenheit. Figure 3 shows that the AI agent was given a prompt to create a transformation step that transformed the temperature readings from Celcius into Fahrenheit.

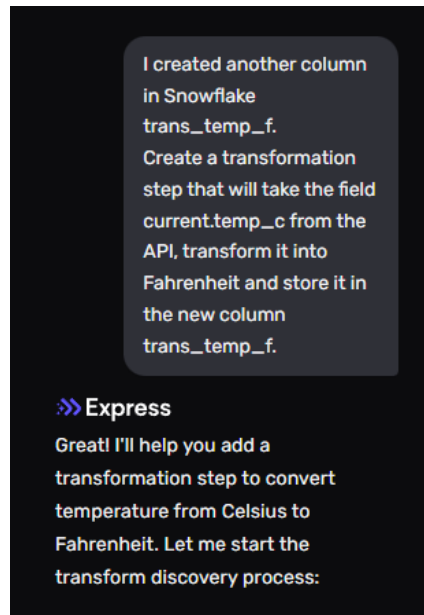


Figure 3: Transformation prompt

This step was done by the AI, implementing the formula of “Fahrenheit = (Celsius  $\times$  9/5) + 32” to generate the transformation as in Figure 4. This transformation logic was implemented by Nexla Express.dev using the Python-based transformation function. Furthermore, before completing the conversion and storing the resulting value into a new field called trans\_temp\_f, the created code verified that the source temperature field existed, and also, records with null values were not processed. This is in order to preserve data integrity and quality.

```
Transform: Code #1 » Python Transform Python

1 def transform(input, metadata, args):
2     current = input.get('current')
3     if current is None:
4         return None
5
6     temp_c = current.get('temp_c')
7     if temp_c is None:
8         return None
9
10    output = input.copy()
11    output['trans_temp_f'] = (temp_c * 9.0 / 5.0) + 32.0
12    return output
13
```

Figure 4: Transformation code by Nexla

Following the transformation's implementation, the AI agent conducted an automated verification procedure to make sure the transformation proceeded smoothly. This verification procedure clarified how crucial it is to validate AI-generated ETL procedures. In order to ensure precise and reliable data integration, some configurations, like destination settings and schema mappings, may need to be manually verified. The pipeline was then turned on to allow weather data from WeatherAPI.com to be continuously extracted, transformed, and loaded into Snowflake, as shown in Figure 5.

|    | TEMPERATURE | HUMIDITY | WEATHER       | TIMESTAMP                | TRANS_TEMP_F |
|----|-------------|----------|---------------|--------------------------|--------------|
| 62 | 21.1        | 56       | Partly cloudy | 2026-06-15T14:30:00.000Z | 69.98        |
| 63 | 21.1        | 56       | Partly cloudy | 2026-06-15T14:30:00.000Z | 69.98        |
| 64 | 21.3        | 56       | Partly cloudy | 2026-06-15T14:45:00.000Z | 70.34        |
| 65 | 22.1        | 53       | Partly cloudy | 2026-06-15T15:15:00.000Z | 71.78        |
| 66 | 22.1        | 53       | Partly cloudy | 2026-06-15T15:15:00.000Z | 71.78        |
| 67 | 21.2        | 49       | Partly cloudy | 2026-06-15T15:45:00.000Z | 70.16        |
| 68 | 21.2        | 49       | Partly cloudy | 2026-06-15T15:45:00.000Z | 70.16        |
| 69 | 21.2        | 49       | Partly cloudy | 2026-06-15T15:45:00.000Z | 70.16        |
| 70 | 21.4        | 49       | Partly cloudy | 2026-06-15T16:00:00.000Z | 70.52        |
| 71 | 21.3        | 49       | Partly cloudy | 2026-06-15T16:15:00.000Z | 70.34        |
| 72 | 21.3        | 49       | Partly cloudy | 2026-06-15T16:15:00.000Z | 70.34        |
| 73 | 22          | 50       | Partly cloudy | 2026-06-15T16:30:00.000Z | 71.6         |
| 74 | 22          | 50       | Partly cloudy | 2026-06-15T16:30:00.000Z | 71.6         |
| 75 | 22.1        | 50       | Partly cloudy | 2026-06-15T17:00:00.000Z | 71.78        |
| 76 | 22.1        | 50       | Partly cloudy | 2026-06-15T17:00:00.000Z | 71.78        |

Figure 5: Data loaded in Snowflake

Figure 6 shows the complete pipeline, including the scheduling, data source, and virtual data product. The purpose of this virtual data product is so that the data can be reused or shared with others in an organization based on their access. Furthermore, this pipeline allows data streaming from [WeatherAPI.com](https://weatherapi.com) every 15 minutes. This WeatherAPI source was configured to poll data every 2 minutes. However, as the Snowflake destination used a micro-batch loading approach, it results in data appearing in the Snowflake every 15 minutes.

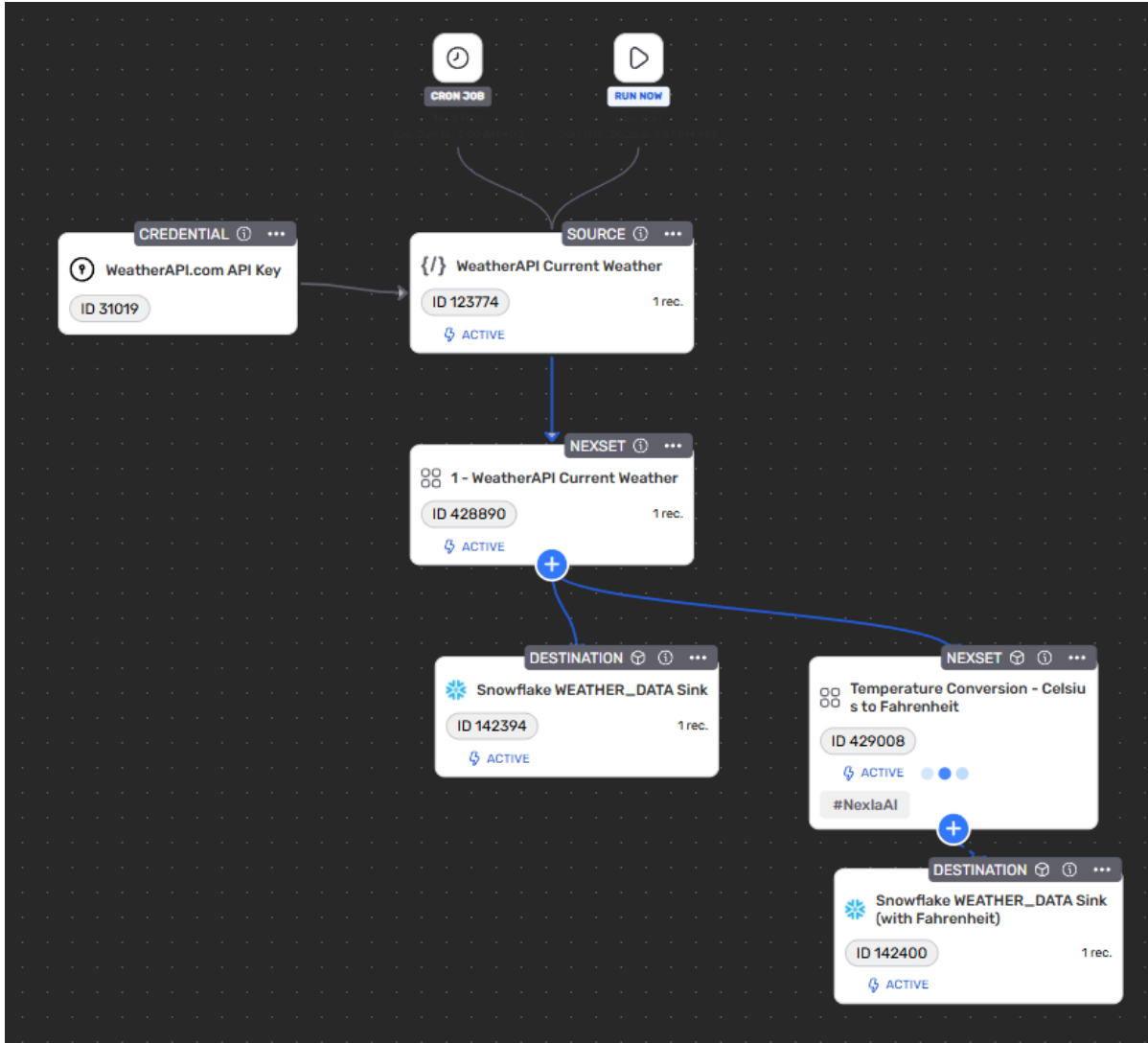


Figure 6: Complete Data Pipeline

## 4.0 Reflection

This section includes individual reflection for this tutorial.

### 4.1 Nurul Asyikin Binti Khairul Anuar

This tutorial provided useful insights into how AI may be used in data engineering specifically to construct ETL pipelines. By using the AI agent in Express.dev, almost all of the pipeline setup process, including connection setup, scheduling, and data transformation, could be completed using a simple prompt. Before this tutorial, there was no awareness that an AI agent could be used to build a complete data pipeline through simple prompts in Express.dev.

This tutorial provided a new perspective on how ETL procedures may be designed in a more automated and interactive way than in a typical manual configuration. It also demonstrated how platforms such as WeatherAPI and Snowflake can be linked more efficiently with AI assistance. However, this tutorial also demonstrated that the created pipeline still has to be verified and checked in order to ensure that it functions properly and produces accurate results.

## **4.2 Lubna Al Haani Binti Radzuan**

The tutorial provided a clearer understanding of AI-assisted data engineering and how it supports data engineering workflows. Through the creation of a simple pipeline using Express.dev, WeatherAPI, and Snowflake, key concepts of AI-assisted data engineering, such as the automation of pipeline development, were explored and understood. Through this tutorial, exposure to new AI tools was gained, particularly Express.dev. The benefit of the tool is that it helps users by simplifying credential setup through guided forms and by automating pipeline configuration based on selected criteria.

This improves work efficiency and reduces manual setup effort in data engineering workflows. As an improvement, error-handling mechanisms can be explored and integrated into the pipeline to enhance its reliability. In conclusion, modern solutions are always evolving, and traditional workflows need a way to be adapted with modern solutions to improve processes and produce more effective and scalable outcomes.

## **4.3 Anis Safiyya Binti Janai**

This tutorial explains in depth AI-assisted data engineering, that is using artificial intelligence to assist in creating ETL pipelines, conducting data conversion, and other tasks. The tutorial demonstrates using natural language prompts to configure data pipelines more efficiently through Express.dev. Compared with traditional manual methods, it significantly enhances the potential of an AI agent in simplifying data engineering workflows, reducing repetitive tasks, and accelerating the development process.

The AI agent used in the tutorial demonstrates its ability to generate pipeline components, establish connections between WeatherAPI and Snowflake, and create conversion logic according to user instructions. However, the tutorial also emphasizes the importance of verifying the output results generated by AI to ensure the accuracy of data processing and the reliability of the pipeline. This shows that AI plays an important role in data engineering, but manual supervision is still crucial.

## REFERENCES

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